Quiz 1:

Question no 1:

You are tasked with developing a multi-process C program that leverages the fork system call to implement parallel data processing. The program should perform the following

1. Take the name of the file from the command line and pass it to parent and child process through pipes.

File Splitting: Read a large text file provided as input and split it into smaller chunks. Each chunk should be processed independently.

Parallel Data Processing: Utilize the fork system call to create a child processe, including the parent process. The child process, as well as the parent process, should be responsible for processing a different chunk of the file in parallel. The processing can include tasks such as:

Counting specific words in the chunk.

Counting the vowels in their respective chunk

Communication and Synchronization: Implement proper communication and synchronization mechanisms between child processes and the parent process. Child processes should report their individual results to the parent process.

Result Consolidation: The parent process should consolidate the results obtained from all child processes. You can choose how to present the consolidated results (e.g., print to a summary file or display on the screen).

Exec: Extend the program by using the exec system call within one of the child processes. This child process should execute another program of your choice (e.g., a different C program or a script). The executed program should perform a specific task (e.g., additional data analysis) on a chunk of data and report its result.

Make sure to provide clear comments and explanations in your code.

Please include the following in your submission:

1 The C program(s) you developed.

2. a makefile to execute your program